

L1a: Course Orientation, Markets, Assets, and the Decision Problem

CHEME 5660 Cornell University

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Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

Today we locate the course, argue that you are **already a participant** in the markets it studies, and end on a question this lecture will not answer.

Today's concepts

- **Locate the course:** how four units connect cash-flow models, uncertainty, contingent claims, and algorithmic decisions.
- **Why you are already in this:** retirement accounts make nearly every working professional a portfolio holder, and the markets involved are roughly a thousand times the daily scale of the industries this college trains you to enter.
- **The decision problem:** an instrument is a schedule of dated payments, and two schedules cannot be ranked without a rule for moving dollars across time.

Note: this lecture has no notebook and no notes packet. These slides are the record.

How This Course Runs

Twenty eight lectures in **four units**, each unit adding one piece of machinery to the same underlying question. Six problem sets, spread across the four units, are where the machinery gets used.

What you need

- **Materials**: each week ships as a zip file on the repository releases page. Cloning works too. Released weeks are frozen: fixes are announced, never silently rewritten.
- **Tools**: Julia 1.12 through juliaup, Anaconda for Jupyter, and VS Code with the Julia extension. One setup step, then the notebooks run.
- **Questions**: ask on the course discussion board rather than by email, so an answer reaches everyone who had the same question.

Policies: [grading](#), [attendance](#), [academic integrity](#), and [accommodations](#) are posted in full

What the Semester Feels Like

The policies are posted and you can read them. What a document cannot tell you is the **rhythm**, so here it is.

- **Every lecture builds one thing and uses it immediately.** Miss a week and the next one is harder, not just longer. This is a course with a spine.
- **You will write code.** Not a lot of it, and not from scratch: the course package does the heavy lifting. But a model you have not run is a model you do not believe.
- **The math is the easy part.** Deciding what to model, and admitting when the model is wrong, is the part that takes the semester.

Questions about any of this: now is the time.

You Already Hold a Portfolio

Roughly 61% of U.S. adults hold a retirement account: a 401(k), a 403(b), or an IRA. In household wealth generation, markets are second only to real estate. (2025)

So the question is not whether to participate. You already do.

- That account holds a **portfolio**: a set of positions, in some proportion, chosen by someone.
- Choosing those proportions is an **allocation decision**, and it is the same decision this course spends four units learning to make.
- Most people holding one have **never consciously made it**. A default was selected for them and they left it alone.

Inaction is an allocation. It is just not an argued one.

Twelve People, Eight Meetings a Year

One committee of twelve meets eight times a year and moves the value of everything in that account.

- **Thursday**: you meet them. What they actually control, what they do not, and why that distinction decides which number you use to value anything.
- **September 15 and 16**: they meet. We are three weeks into the course by then, and we come back to it in week four, with the tools to say what the decision did.

That is the whole teaser. Sixty seconds, and we move on.

Link: [FOMC calendar and meeting materials](#)

The Decision Problem

Three kinds of asset, one question. You observe something, you do something, and something comes back. The three words for those are **state**, **action**, and **reward**.

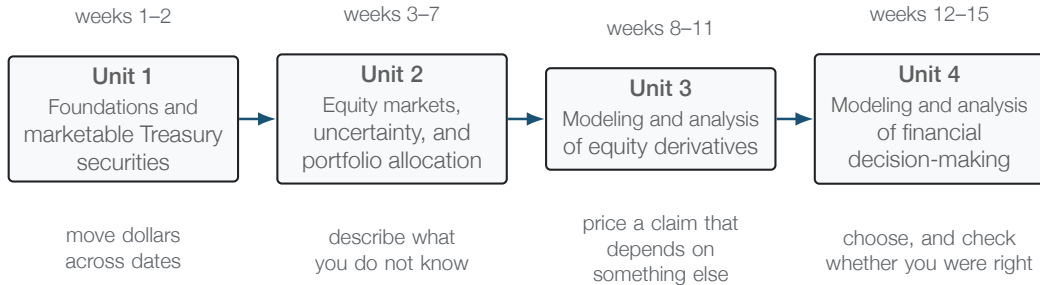


The loop is easy to draw and hard to close. Closing it needs a way to compare payments that arrive on different dates, a way to describe what you do not know, a way to price a claim that depends on something else, and a rule for choosing. Those are the four units.

Careful: the picture is not the problem. Every arrow in it is a modeling choice.

Four Units

Each unit adds exactly one capability, and the next one cannot be built without it.



We start at the left today, and we do not get past it until Thursday.

Summary

This lecture located the course, argued that you are already a market participant, and read five financial promises as schedules of dated payments.

Key takeaways

- **You are already a portfolio holder.** Retirement accounts make market participation the default, so the allocation question is one nearly everyone answers, most of them by inaction.
- **An instrument is a schedule.** A bill, a note, a bond, a corporate bond, and an inflation-indexed security differ in their dated cash flows, not in kind: the product names are labels for those schedules.
- **A claim is only as good as the rung it survived.** Theory, backtest, forward test, and real money each see different data, and each is the only stage able to falsify its own claim.

On Thursday we build the rule that lets us rank those five schedules.